NICHOLAS SOBCHAK

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Portfolio | LinkedIn | GitHub | U.S. Citizen

EDUCATION

University of Maryland, Baltimore County (UMBC) Expected May 2027
Bachelor of Science in Computer Science
Relevant coursework: Operating Systems, Neurosymbolic AI, Database Management Systems, Data Visualization

Carroll Community College May 2025
Associate of Arts in Computer Science

TECHNICAL SKILLS

Languages: C++17, Python, Java, SQL, TypeScript, JavaScript
Systems & Tools: Linux, Git, Docker, CMake, GitHub Actions, Valgrind
Testing: Catch2, pytest, JUnit
Backend & Data: Spring Boot, JPA, PostgreSQL, SQLite, DynamoDB
Cloud: AWS Lambda, API Gateway, Cognito, SAM, CloudFormation, boto3
Frontend: Angular, React, HTML, CSS

EXPERIENCE

Founding Software Developer – ZipTip Dec 2025 – Present
Remote / Westminster, MD

- Designed a serverless scan API with AWS Lambda, API Gateway, and DynamoDB, implementing authentication, custom-domain routing, and ID-to-account redirects for a real-time tipping platform.
- Built AWS SAM CI/CD pipelines with OIDC authentication, automated Route 53 DNS updates, and rollback support for development and production environments.
- Wrote pytest unit and integration tests to validate backend services and support automated AWS SAM deployments.
- Refactored Angular services and typed API clients; introduced shared linting tooling across the codebase.

TECHNICAL PROJECTS

QuickQuill – *C++17, Java, Spring Boot, SQLite, PostgreSQL, Docker*

- Built a trie-based C++ dictionary engine supporting lookup, spell-check suggestions, and autocomplete across a 1.28M-word dictionary.
- Used `std::mutex` for thread-safe LRU caching and `thread_local` SQLite connections for concurrent lookups; managed trie nodes and database handles using `std::unique_ptr`.
- Integrated the native engine with a Spring Boot REST API through Java's Foreign Function & Memory API and an `extern "C"` interface.
- Tested data structures, dictionary operations, and concurrent lookups with Catch2; packaged C++ and Java services in a multi-stage Docker build.

On The Money – *C++17, Java, Spring Boot, PostgreSQL, React, Docker*

- Integrated a C++ Monte Carlo engine with a Spring Boot REST API to run 10,000 simulations per request for retirement and portfolio growth projections.
- Wrote Catch2 tests for percentile calculations, trajectory consistency, and simulation edge cases.
- Integrated Plaid Link for bank-account connections and transaction synchronization; deployed a Dockerized Java API and React PWA backed by PostgreSQL.

ACTIVITIES

UMBC Ice Hockey – Player
Community Orchestra – Section Leader